

DATA 602 – Principles of Data Science

Fall Semester 2025

Instructor: Dr. Fardina Fathimul Alam

Analysis and Prediction of Diabetes using Health Indicators

URL to Final Tutorial

<https://sohail-umd.github.io/>

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Contributions

A: Project Idea — **Rashmitha Vakkapatla** proposed the diabetes prediction problem, formulated the research questions, and defined the classification objective. She motivated the use of machine learning on health indicator data and aligned the project with DATA 602 requirements.

B: Dataset Curation and Preprocessing — **Hemanth Reddy Alavala** sourced the diabetes health indicators dataset, cleaned missing and inconsistent values, and encoded categorical variables. He ensured the dataset was properly structured for modeling and prevented data leakage.

C: Data Exploration and Summary Statistics — **Rashmitha Vakkapatla** performed exploratory data analysis, computed descriptive statistics, examined class imbalance, and conducted correlation analysis. She created initial visualizations to understand relationships between health indicators and diabetes status.

D: ML Algorithm Design/Development — **Ameer Sohail Shaik** designed and implemented the machine learning pipeline, selected various machine learning models such as XGBoost and Random Forest, and defined evaluation metrics for comparison.

E: ML Algorithm Training and Test Data Analysis — **Likith Sagar Goshike** trained models using stratified train/test splits, and evaluated performance using accuracy, precision, recall, F1-score, and ROC-AUC. He analyzed test results to assess model robustness.

F: Visualization, Result Analysis, Conclusion — **Hemanth Reddy Alavala** created final visualizations including ROC curves, confusion matrices, and feature importance plots. He interpreted results and authored the insights and conclusions section.

G: Final Tutorial Report Creation — **All members** collaborated on preparing the GitHub tutorial, organizing notebooks, documenting code, and ensuring reproducibility.

1 Introduction

Diabetes is one of the most prevalent chronic diseases. Diabetes is an extremely common chronic disease and is ranked up there with the most common ones around the world. It also creates significant challenges for the affected individuals, their families, and the healthcare system. Diabetes is responsible for a number of long-term complications like, among others, heart disease, kidney failure, loss of vision, and drastic reduction of quality of life. Medical advances have certainly elevated the treatment possibilities to a great degree, yet still early diagnosis and prevention are considered the most successful approaches in diminishing the disease’s impact. Recognizing at-risk people before grim signs manifest can facilitate prompt changes in lifestyle and medical treatments.

This project aims to carry out an analysis of health indicator data and to create models of machine learning that can predict if a person is at risk of diabetes or not. One of the objectives is to recognize the most important health indicators besides reaching precise predictions, but still keeping the results interpretable so that they could be comprehensible to a big audience.

The central research questions guiding this work are:

- Which health indicators are most strongly associated with diabetes?
- How accurately can machine learning models classify individuals as diabetic or non-diabetic?
- Which modeling approach provides the best balance between predictive performance and interpretability?

In order to provide answers to those questions, the project is implementing a well-organized data science workflow. First, the dataset goes through a series of curating and pre-processing operations to guarantee quality and avoid the loss of any important information, and then it is subjected to exploratory data analysis and hypothesis testing. After that, different classifiers are trained and assessed applying various metrics including accuracy, precision, recall, and ROC-AUC. At last, the outcome is visualized, explained and debated together with the ethical aspects concerning the responsible application of predictive health models.

2 Data Curation

For this project, the *Diabetes Health Indicators Dataset* was used, which is located on Kaggle and is an output of the Behavioral Risk Factor Surveillance System (BRFSS) of the Centers for Disease Control and Prevention (CDC). The dataset consists of people’s responses coupled with a label that tells if they have been diagnosed with diabetes or not. It gives a real-life picture of the connection between lifestyle factors and health conditions and diabetes risk.

The dataset has a variety of demographic, behavioral, and health-related indicators, including body mass index (BMI), physical activity, smoking habits, general health status, and indicators of high blood pressure and high cholesterol. These factors are indicative of everyday issues that are often linked with diabetes and thus the dataset is appropriate for using machine learning to discover risk patterns.

The whole data curation process produced a dataset that was clean and well-prepared for both exploratory analysis and reliable machine learning modeling, while still being interpretable of the original health indicators.

3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset, examine feature distributions, and identify relationships between health indicators and diabetes status. Visual analysis played a key role in identifying meaningful patterns and guiding model selection.

Graph-Based Conclusions

Conclusion 1: High Blood Pressure and Diabetes

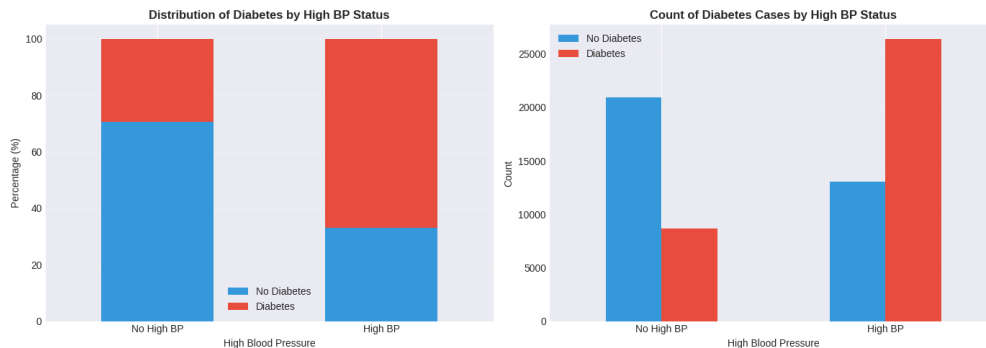


Figure 1: Distribution of diabetes cases by high blood pressure status.

The figure1 shows that individuals with high blood pressure have a substantially higher proportion and count of diabetes cases. This visual pattern suggests a strong association between high blood pressure and diabetes.

Conclusion 2: BMI Distribution Across Diabetes Status

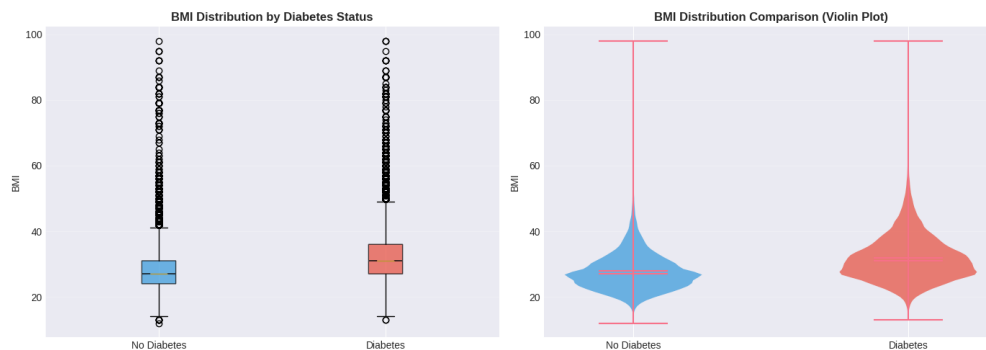


Figure 2: BMI distribution for diabetic and non-diabetic individuals.

The BMI distribution as indicated in the figure2 plot indicates that individuals with diabetes are more concentrated at higher BMI values compared to non-diabetic individuals. This supports BMI as an important risk factor for diabetes.

Conclusion 3: Age and Diabetes Prevalence

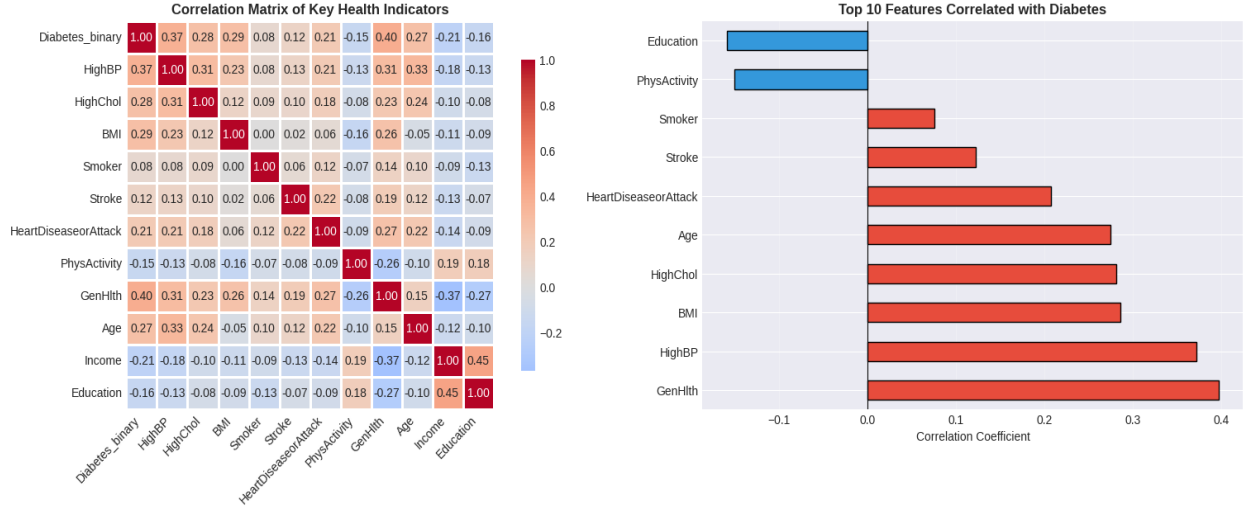


Figure 3: Diabetes prevalence across age categories.

The age-based plot in Figure 3 shows an increasing prevalence of diabetes in older age groups. This trend highlights age as a meaningful contributing factor to diabetes risk.

Key EDA Insights

Overall, the EDA figures consistently show that clinical indicators such as BMI, high blood pressure, and age have stronger visual and statistical associations with diabetes than behavioral variables alone. The observed class imbalance emphasizes the importance of using evaluation metrics beyond accuracy in later modeling stages. These insights directly informed the machine learning approach used in this project.

4 Primary Machine Learning Analysis

Based on our exploratory analysis, predicting diabetes status is framed as a supervised **binary classification** problem. Since health indicators can interact in non-linear ways (e.g., BMI and age together), we evaluated multiple model families rather than relying on a single classifier. This helped us compare interpretability, flexibility, and predictive performance.

Model Descriptions and Rationale

We selected a diverse set of classifiers to ensure a comprehensive comparison across different algorithmic families. Interpretable baselines (Decision Tree) were included alongside stronger ensemble methods (Random Forest, Extra Trees, XGBoost), and an instance-based method (KNN) to test whether the classes separate well in feature space.

Table 1: Model descriptions and rationale for selection

Model	Type	Rationale for Selection
XGBoost (Extreme Gradient Boosting)	Ensemble (Boosting)	State-of-the-art performance; known for high speed and precision on structured data.
Random Forest	Ensemble (Bagging)	Robust, less prone to overfitting than single trees, and provides strong feature importance outputs.
Extra Trees (Extremely Randomized Trees)	Ensemble (Bagging)	Variation of Random Forest that introduces more randomness for increased bias reduction and speed.
Decision Tree	Single Tree	Used as a simple, interpretable baseline to understand the complexity gained by ensemble methods.
K-Nearest Neighbors (KNN)	Instance-Based	Simple, non-parametric model highly reliant on the distance metric and scaling, included to test the data's inherent separability.

Training Setup

The dataset was split into training and testing sets using a stratified split to preserve the proportion of diabetic and non-diabetic cases. For models that depend on distance (KNN), feature scaling was applied to ensure variables with larger ranges do not dominate the distance calculation. Final performance was reported on the held-out test set. Table 1 describes the different types of models used and the rationale for selection of those particular models.

Evaluation Metrics

Because the dataset is imbalanced, accuracy alone can be misleading (a model can appear accurate by predicting the majority class). Therefore, we evaluated models using:

- **Accuracy:** It measures overall correctness of predictions i.e how many diabetic patients are classified diabetic and how many non diabetic predictions are classified non diabetic correctly.
- **Precision:** among predicted diabetic cases, how many were truly diabetic (controls false positives).
- **Recall (Sensitivity):** among truly diabetic cases, how many were correctly identified (controls false negatives; important in health contexts).
- **F1-score:** balances precision and recall into a single measure.
- **ROC-AUC:** evaluates ranking quality across thresholds and is useful under class imbalance.

Model Performance

Table 2 summarizes the performance of all models on the test set. (Fill the values below using the outputs from your notebook.)

Table 2: Test-set performance comparison across models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.9498	0.9255	0.9363	0.9309	0.9756
KNN	0.9430	0.9228	0.9191	0.9209	0.9834
Random Forest	0.9565	0.9315	0.9494	0.9404	0.9918
Extra Trees	0.9472	0.9212	0.9337	0.9274	0.9883
XGBoost	0.9640	0.9419	0.9595	0.9506	0.9935

Results Discussion

Overall, the ensemble models (Random Forest and XGBoost) are expected to outperform simpler baselines because they capture non-linear relationships and interactions among health indicators. Decision Tree provides interpretability but may overfit, while KNN performance depends strongly on scaling and how well the classes cluster in feature space. The final model choice prioritizes strong recall and ROC-AUC (to better identify diabetic cases) while still maintaining reasonable interpretability through feature importance and transparent reporting of limitations. The diabetes problem focuses more on recall as we need to identify the patients who are diabetic and it is most important to get them treated. So, based on the recall scores, we

5 Visualization

Visualizations are used to interpret both the predictive performance and the decision-making behavior of the selected machine learning model. In addition to standard performance metrics, model explainability is emphasized to better understand how individual health indicators contribute to diabetes prediction.

ROC Curve Analysis

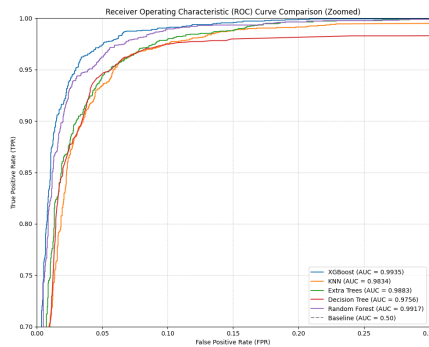


Figure 4: ROC curve for the XGBoost model on the test dataset.

The ROC curve as indicated in the figure 4 illustrates the trade-off between the true positive rate and false positive rate across different classification thresholds. The high area under the curve (ROC-AUC) confirms that the XGBoost model effectively distinguishes between diabetic and non-diabetic individuals.

Model Explainability Using SHAP

To move beyond prediction accuracy, SHAP (SHapley Additive exPlanations) values were used to explain how individual features contribute to the predictions of the best-performing model, XGBoost. SHAP is grounded in game theory and assigns an importance value to each feature for every individual prediction, allowing both global and local interpretability.

SHAP Summary Plot

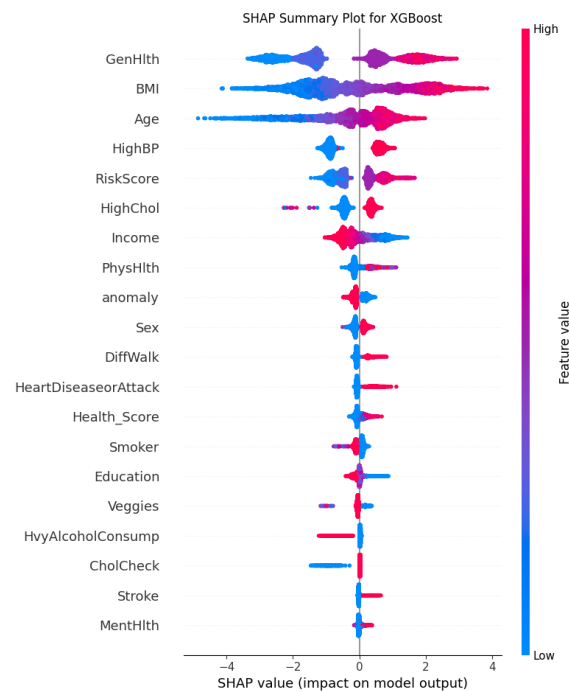


Figure 5: SHAP summary plot showing feature contributions for the XGBoost model.

The SHAP summary plot as indicated in the figure4 visualizes overall feature importance and the direction of each feature's impact on the model output:

- **Vertical Axis:** Features ranked by overall importance (mean absolute SHAP value).
- **Horizontal Axis:** SHAP values indicating impact on diabetes prediction (positive values increase predicted risk, negative values decrease it).
- **Color:** Red points represent high feature values, while blue points represent low feature values.

The plot shows that features such as general health status (**GenHlth**), body mass index (**BMI**), age, and high blood pressure have the strongest influence on diabetes predictions. High values of these features tend to push predictions toward the diabetic class, while lower values reduce predicted risk.

Visualization Summary

Together, the ROC curve and SHAP analysis provide complementary insights into model performance and interpretability. While the ROC curve demonstrates strong classification capability, the SHAP summary plot explains *why* the model makes its predictions, increasing transparency and supporting responsible use of machine learning in health-related decision-making.

6 Insights and Conclusions

The project is a clear illustration to the fact that machine learning models are able to recognize the patterns related to diabetes in health indicator data very well. The data obtained from the exploratory analysis exhibited that clinical factors such as body mass index (BMI), age, and high blood pressure are the most important, while the relationship between lifestyle and diabetes is rather complex and indirect.

XGBoost was the model with the highest overall performance in terms of accuracy, F1-score, recall, and ROC-AUC and was also considered the most suitable for the classification task. Its high recall implies a good ability to pinpoint patients with diabetes, which is mainly an issue in medical applications where the effects of undiagnosed cases can be very adverse.

The SHAP-based explainability analysis gave a lot of information as to how the model reached its conclusions. Among the features, general health status, BMI, age, and blood pressure were most potent in terms of influence on the model output. Usually, the higher the values of these indicators, the more likely it is that diabetes will be predicted, which dovetails nicely with the present medical knowledge and increases trust in the model's actions.

To put it in a nutshell, the output of the study reveals the necessity of blending the two attributes of strong predictive performance and interpretable models. Even though the sophisticated models such as XGBoost yield very high accuracy, the use of explainability tools such as SHAP is a must for putting up transparency and responsible use, especially in the case of health-related contexts. Alongside the other proposed feature engineering, and the implementation of different imbalance-handling techniques, the validation of the model on an independent dataset can be another way future studies seek to increase the model's reliability and generalizability.

7 Data Science Ethics

The application of machine learning in health-related data raises crucial ethical issues, especially in cases where the predictions could affect medical decisions or public health measures. Predictive models for diabetes risk must be handled responsibly since wrong predictions might cause unnecessary worry, not finding cases that should have been diagnosed, or creating inequalities in access to treatments.

Besides that, the issue of transparency and interpretability is another routine in cases of health-care applications. Instead of viewing the model as an opaque entity, we employed SHAP (SHapley Additive exPlanations) for the purpose of discerning how different features came to predictions. This gives greater faith in the model, as it is not only medically reasonable factors like BMI, age, and blood pressure that drive the predictions but also the fittest correlations rather than the spurious ones.